PROPERTIES OF THE FUNCTOR FROM QUASI-COHERENT MODULES TO MODULES ON A BIG OR SMALL SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries the author's own footnote warning that part (10) is misleading in view of part (6); the footnote is inside the counted statement span and is reproduced verbatim.

Editorial note. The counted spans carry 10 navigation stubs, one per cited result; none of them replaces a load-bearing step, since the counterexamples and the final reduction are written out in the span. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The functor sending a quasi-coherent module to the module it represents on a tau-site. Same chapter, Proposition proposition-equivalence-quasi-coherent, invoked at proof lines 3310-3311 and named in the statement. Supplies the functors whose properties are listed and part (1).
- The construction as a pullback of ringed sites and the exactness properties of such pullbacks. Same chapter, Remark remark-change-topologies-ringed-sites and Modules on Sites, Lemma sites-modules-lemma-exactness-pushforward-pullback, invoked at proof lines 3317-3322. Supplies commutation with colimits and hence right exactness.
- Etale morphisms are flat, and pullback along flat morphisms of ringed sites commutes with internal hom out of a finitely presented module. Morphisms, Lemma morphisms-lemma-etale-flat and Modules on Sites, Lemmas sites-modules-lemma-tensor-product-pullback and sites-modules-lemma-pullback-internal-hom, invoked at proof lines 3330-3362. Supplies parts (5), (7) and (8).
- Cohomology of a quasi-coherent module agrees on the small and big sites on affines. Same chapter, Proposition proposition-same-cohomology-quasi-coherent, invoked at proof lines 3385-3388. Supplies the vanishing of the first cohomology used in part (10).

Source reference. Original definition 058I: sources/stacks-chow/tags.txt, source lines 6781–6781. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GNA: sources/stacks-descent/pinned-original-source.tex, source lines 3221–3221. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GS: sources/stacks-chow/tags.txt, source lines 3239–3239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03DX: sources/stacks-descent/pinned-original-source.tex, source lines 2716–2716. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03DW: sources/stacks-descent/pinned-original-source.tex, source lines 2990–2990. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 03FH: sources/stacks-descent/pinned-original-source.tex, source lines 2587–2587. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 01LA: sources/stacks-chow/tags.txt, source lines 2079–2079. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03DC: sources/stacks-chow/tags.txt, source lines 4415–4415. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GN2: sources/stacks-chow/tags.txt, source lines 20777–20777. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03EL: sources/stacks-chow/tags.txt, source lines 4460–4460. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. The functors*

$$QCoh(\mathcal{O}_S) \longrightarrow Mod((Sch/S)_\tau, \mathcal{O}) \quad \text{and} \quad QCoh(\mathcal{O}_S) \longrightarrow Mod(S_\tau, \mathcal{O})$$

defined by the rule $\mathcal{F} \mapsto \mathcal{F}^a$ seen in Proposition 03DX are

1. *fully faithful,*
2. *commmute with direct sums,*
3. *commmute with colimits,*
4. *right exact,*

(5) *exact as a functor* $QCoh(\mathcal{O}_S) \to Mod(S_{étale}, \mathcal{O})$,
(6) **not** *exact as a functor* $QCoh(\mathcal{O}_S) \to Mod((Sch/S)_\tau, \mathcal{O})$ *in general,*
(7) *given two quasi-coherent* $\mathcal{O}_S$*-modules* $\mathcal{F}$, $\mathcal{G}$ *we have* $(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{G})^a = \mathcal{F}^a \otimes_{\mathcal{O}} \mathcal{G}^a$,
(8) *if* $\tau = étale$ *or* $\tau = Zariski$, *given two quasi-coherent* $\mathcal{O}_S$*-modules* $\mathcal{F}$, $\mathcal{G}$ *such that* $\mathcal{F}$ *is of finite presentation we have* $(\mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{G}))^a = \mathcal{H}om_{\mathcal{O}}(\mathcal{F}^a, \mathcal{G}^a)$ *in* $Mod(S_\tau, \mathcal{O})$,
(9) *given two quasi-coherent* $\mathcal{O}_S$*-modules* $\mathcal{F}$, $\mathcal{G}$ *we do* **not** *have* $(\mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{G}))^a = \mathcal{H}om_{\mathcal{O}}(\mathcal{F}^a, \mathcal{G}^a)$ *in* $Mod((Sch/S)_\tau, \mathcal{O})$ *in general even if* $\mathcal{F}$ *is of finite presentation, and*
(10) *given a short exact sequence* $0 \to \mathcal{F}_1^a \to \mathcal{E} \to \mathcal{F}_2^a \to 0$ *of* $\mathcal{O}$*-modules then* $\mathcal{E}$ *is quasi-coherent*[1]*, i.e.,* $\mathcal{E}$ *is in the essential image of the functor.*

Proof. Part (1) we saw in Proposition 03DX.

We have seen in Schemes, Section 01LA that a colimit of quasi-coherent sheaves on a scheme is a quasi-coherent sheaf. Moreover, in Remark 03FH we saw that $\mathcal{F} \mapsto \mathcal{F}^a$ is the pullback functor for a morphism of ringed sites, hence commutes with all colimits, see Modules on Sites, Lemma 03DC. Thus (3) and its special case (2) hold.

This also shows that the functor is right exact (i.e., commutes with finite colimits), hence (4).

The functor $QCoh(\mathcal{O}_S) \to Mod(S_{étale}, \mathcal{O})$, $\mathcal{F} \mapsto \mathcal{F}^a$ is left exact because an étale morphism is flat, see Morphisms, Lemma 02GS. This proves (5).

To see (6), suppose that $S = \mathrm{Spec}(\mathbf{Z})$. Then $2 : \mathcal{O}_S \to \mathcal{O}_S$ is injective but the associated map of $\mathcal{O}$-modules on $(Sch/S)_\tau$ isn't injective because $2 : \mathbf{F}_2 \to \mathbf{F}_2$ isn't injective and $\mathrm{Spec}(\mathbf{F}_2)$ is an object of $(Sch/S)_\tau$.

Part (7) holds because, as mentioned above, the functor $\mathcal{F} \mapsto \mathcal{F}^a$ is the pullback functor for a morphism of ringed sites and such commute with tensor products by Modules on Sites, Lemma 03EL.

Part (8) is obvious if $\tau = Zariski$ because the category of $\mathcal{O}$-modules on S_{Zar} is the same as the category of $\mathcal{O}_S$-modules on the topological space S. If $\tau = étale$ then (8) holds because, as mentioned above, the functor $\mathcal{F} \mapsto \mathcal{F}^a$ is the pullback functor for the flat morphism of ringed sites $(S_{étale}, \mathcal{O}) \to (S_{Zar}, \mathcal{O}_S)$, see Lemma 0GNA. Pullback by flat morphisms of ringed sites commutes with taking internal hom out of a finitely presented module by Modules on Sites, Lemma 0GN2.

To see (9), suppose that $S = \mathrm{Spec}(\mathbf{Z})$. Let $\mathcal{F} = \mathrm{Coker}(2 : \mathcal{O}_S \to \mathcal{O}_S)$ and $\mathcal{G} = \mathcal{O}_S$. Then $\mathcal{F}^a = \mathrm{Coker}(2 : \mathcal{O} \to \mathcal{O})$ and $\mathcal{G}^a = \mathcal{O}$. Hence $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}^a, \mathcal{G}^a) = \mathcal{O}[2]$ is equal to the 2-torsion in $\mathcal{O}$, which is not zero, see proof of (6). On the other hand, the module $\mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{G})$ is zero.

Proof of (10). Let $0 \to \mathcal{F}_1^a \to \mathcal{E} \to \mathcal{F}_2^a \to 0$ be a short exact sequence of $\mathcal{O}$-modules with $\mathcal{F}_1$ and $\mathcal{F}_2$ quasi-coherent on S. Consider the restriction

$$0 \to \mathcal{F}_1 \to \mathcal{E}|_{S_{Zar}} \to \mathcal{F}_2$$

to S_{Zar}. By Proposition 03DW we see that on any affine $U \subset S$ we have $H^1(U, \mathcal{F}_1^a) = H^1(U, \mathcal{F}_1) = 0$. Hence the sequence above is also exact on the right. By Schemes,

[1]Warning: This is misleading. See part (6).

Section 01LA we conclude that $\mathcal{F} = \mathcal{E}|_{S_{Zar}}$ is quasi-coherent. Thus we obtain a commutative diagram

$$\begin{array}{ccccccccc} & & & & \mathcal{F}_1^a & \longrightarrow & \mathcal{F}^a & \longrightarrow & \mathcal{F}_2^a & \longrightarrow & 0 \\ & & & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{F}_1^a & \longrightarrow & \mathcal{E} & \longrightarrow & \mathcal{F}_2^a & \longrightarrow & 0 & & \end{array}$$

To finish the proof it suffices to show that the top row is also right exact. To do this, denote once more $U = \mathrm{Spec}(A) \subset S$ an affine open of S. We have seen above that $0 \to \mathcal{F}_1(U) \to \mathcal{E}(U) \to \mathcal{F}_2(U) \to 0$ is exact. For any affine scheme V/U, $V = \mathrm{Spec}(B)$ the map $\mathcal{F}_1^a(V) \to \mathcal{E}(V)$ is injective. We have $\mathcal{F}_1^a(V) = \mathcal{F}_1(U) \otimes_A B$ by definition. The injection $\mathcal{F}_1^a(V) \to \mathcal{E}(V)$ factors as

$$\mathcal{F}_1(U) \otimes_A B \to \mathcal{E}(U) \otimes_A B \to \mathcal{E}(V)$$

Considering A-algebras B of the form $B = A \oplus M$ we see that $\mathcal{F}_1(U) \to \mathcal{E}(U)$ is universally injective (see Algebra, Definition 058I). Since $\mathcal{E}(U) = \mathcal{F}(U)$ we conclude that $\mathcal{F}_1 \to \mathcal{F}$ remains injective after any base change, or equivalently that $\mathcal{F}_1^a \to \mathcal{F}^a$ is injective. $\square$